

UltraShip | AI Engineer Skill Test

Format: Take-home exercise (Parts 1–3) + 45-minute live review (Part 4)

Time expectation: 3–4 hours total. We respect your time — do not over-polish. We care about your reasoning and judgment more than production-grade code.

Deliverable: 1. A GitHub repo (or zip) with your code, 2. A short README, 3. Written answer to design question

About the role

UltraShip is a Transportation Management System (TMS) for non-asset freight brokers and 3PLs. Our AI work centers on automating the messy, document-heavy parts of freight brokerage: rate confirmations, bills of lading, carrier packets, load board data, and email negotiations. You'll be building LLM-powered features that brokers rely on to move real freight — so accuracy, reliability, and graceful failure matter more than demos.

Part 1 — Document Extraction (core exercise, ~90 min)

Freight brokers receive **rate confirmations** as unstructured PDFs and emails, in hundreds of different formats depending on the shipper or carrier.

Task: Build a small pipeline (Python or TypeScript) that takes the raw text of a rate confirmation and extracts a structured JSON object:

```
{
  "load_id": string | null,
  "origin": { "city": string, "state": string, "zip": string | null },
  "destination": { "city": string, "state": string, "zip": string | null },
  "pickup_date": "YYYY-MM-DD" | null,
  "delivery_date": "YYYY-MM-DD" | null,
  "equipment_type": "van" | "reefer" | "flatbed" | "other" | null,
  "line_haul_rate": number | null,
  "fuel_surcharge": number | null,
  "total_rate": number | null,
  "weight_lbs": number | null,
  "commodity": string | null,
  "confidence": "high" | "medium" | "low"
}
```

Three sample rate confirmations are provided in the `samples` folder.

Requirements:

1. Use an LLM API of your choice (Anthropic, OpenAI, or local model).
2. Enforce the schema — malformed model output should never crash the pipeline or silently produce bad data.
3. Implement the confidence field with real logic, not vibes. Explain in the README how you decide when extraction is trustworthy enough to auto-populate a load vs. flag for human review.
4. Handle at least these failure cases explicitly: missing fields, conflicting totals (line haul + fuel $\neq$ total), and dates written ambiguously (e.g., "3/4/26").

What we're evaluating: prompt design, schema enforcement strategy (structured outputs, retries, validation), and your judgment about when *not* to trust the model.

Part 2 — Evaluation & Reliability (~45 min)

You shipped Part 1. Now the VP of Ops asks: *“How do we know this thing works, and how will we know if it breaks?”*

Task: Write a short document (1 page max) answering:

1. How would you build an eval set for this extraction task? What's your ground truth, how many examples, and how do you score partial correctness (e.g., right rate, wrong date)?
2. Which metric matters most for a broker, and why? (Hint: think about the cost asymmetry — a wrong rate that auto-books is worse than a flagged-for-review load.)
3. How would you detect drift or regressions in production — e.g., a new shipper's document format tanks accuracy, or a model provider update changes behavior?
4. Where would you put a human in the loop, and what would that UI moment look like?

What we're evaluating: whether you think like an engineer shipping AI to non-technical users with real money on the line, not a benchmark chaser.

Part 3 — System Design (written, ~45 min)

UltraShip integrates with DAT and Truckstop (load boards), Highway (carrier identity/fraud), and QuickBooks (accounting).

Scenario: We want to build **“Carrier Match”** — when a broker posts a load, the system suggests the 5 best carriers to call first, ranked by likelihood of covering the load at a good rate.

Task: Write a design sketch (1–2 pages, diagrams welcome) covering:

1. What data sources you'd use and what features you'd derive (lane history, equipment, past rates, fraud signals from Highway, payment history from QuickBooks, etc.).
2. Where LLMs genuinely add value here vs. where classical ML or plain heuristics are the right tool. Be honest — “LLM everything” is a wrong answer.
3. How you'd handle the cold-start problem for a new brokerage with no history.
4. Latency, cost, and caching considerations if this runs on every load posting.

What we're evaluating: architectural judgment, tool selection honesty, and freight-domain reasoning (we don't expect deep freight expertise — we expect you to reason well about a domain you're new to).

CARRIER RATE & LOAD CONFIRMATION



9456 Laguna Springs Drive, Elk Grove, CA, USA, Elk Grove
California 95758-7987
Mailing Address: 9456 Laguna Springs Drive, Elk Grove, CA, USA
Phone: 12312312333
Fax: +12345678901
After hour Contact: test@gmail.com

DETAIL	VALUE
Dispatcher	Test User Ultraship
Phone	+11111111111
Email	testuser@ultraship.ai
Reference ID	LD64392
Booked On	28-Jul-2026
Rate Con Date	28-Jul-2026
Pickup Date	30-Jul-2026

Carrier Details

CARRIER	MC #	PHONE	EQUIPMENT	AGREED AMOUNT (USD)	SIZE (IN FEET)
10314803 CANADA INC	MC1146347	(905) 670-6882	Flatbed	\$50.00 USD	10

Stops

#	Stops	Date/Time	Commodity			
			S. No	Commodity	Weight	Quantity
1	Pickup test automation pickup Illinois State Police, 100 W Randolph St, Chicago, IL 60601, USA PO/Container No 799917 Load Type FTL	Appt Type appointment Shipping Date & Time 07/30/2026 Appointment Time -				
			1	Ceramics	182	07
			Description Automation pickup description			
2	Drop test automation drop New York University Jeffrey S. Gould Welcome Center, 50 W 4th Street, New York, NY 10012, USA PO/Container No 494224 Load Type FTL	Appt Type fcfs Delivery Date & Time 08/01/2026 Appointment Time -				
			1	Ceramics	182	07
			Description This is automation drop location			

Rate Breakdown

Base Carrier Rate	50.00 USD
Total	50.00 USD

Shipper Instructions (Sample)

Please check in with the shipping office upon arrival. Freight must be loaded according to the scheduled appointment time. Contact the shipper immediately if any delays are expected.

Carrier Instructions (Sample)

Driver must call dispatch after pickup and provide status updates every morning. Submit signed POD within 24 hours of delivery to avoid payment delays

Driver Details

DRIVER NAME	CELL NO	TRUCK NO	TRAILER NO
Playwright Driver	(415) 555-1886	33825	65962

Accepted by	Date	Signature
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CARRIER RATE & LOAD CONFIRMATION



9456 Laguna Springs Drive, Elk Grove, CA, USA, Elk Grove
California 95758-7987

Mailing Address: 9456 Laguna Springs Drive, Elk Grove, CA, USA

Phone: 12312312333

Fax: +12345678901

After hour Contact: test@gmail.com

DETAIL	VALUE
Dispatcher	Test User Ultraship
Phone	+11111111111
Email	testuser@ultraship.ai
Reference ID	LD64408
Booked On	28-Jul-2026
Rate Con Date	28-Jul-2026
Pickup Date	03-Aug-2026

Carrier Details

CARRIER	MC #	PHONE	EQUIPMENT	AGREED AMOUNT (USD)	SIZE (IN FEET)
10314803 CANADA INC	MC1146347	(905) 670-6882	Flatbed	\$700.00 USD	53

Stops

#	Stops	Date/Time	Commodity			
1	Pickup test-r Miami International Airport (MIA), Northwest 42nd Avenue, Miami, FL, USA PO/Container No - Load Type FTL	Appt Type appointment Shipping Date & Time 07/28/2026 Appointment Time -	S. No	Commodity	Weight	Quantity
			1	Ceramics	-	g
			Description -			
2	Pickup test automation pickup Illinois State Police, 100 W Randolph St, Chicago, IL 60601, USA PO/Container No - Load Type FTL	Appt Type appointment Shipping Date & Time 08/03/2026 Appointment Time -	S. No	Commodity	Weight	Quantity
			1	Commodity_t	-	55
			Description -			
3	Drop test122 Hertz Car Rental - San Jose - San Jose Mineta International Airport (SJC), Airport Boulevard, San Jose, CA, USA PO/Container No - Load Type FTL	Appt Type appointment Delivery Date & Time 08/05/2026 Appointment Time -	S. No	Commodity	Weight	Quantity
			1	Commodity_t	-	55
			2	Ceramics	-	g
			Description -			

Rate Breakdown

Base Carrier Rate	500.00 USD
Carrier Charge	200.00 USD
Total	700.00 USD

Shipper Instructions (Sample)

Please check in with the shipping office upon arrival. Freight must be loaded according to the scheduled appointment time. Contact the shipper immediately if any delays are expected.

Carrier Instructions (Sample)

Driver must call dispatch after pickup and provide status updates every morning. Submit signed POD within 24 hours of delivery to avoid payment delays

Driver Details

DRIVER NAME	CELL NO	TRUCK NO	TRAILER NO
John Driver	(415) 555-2671	12345	67890

Accepted by	Date	Signature
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CARRIER RATE & LOAD CONFIRMATION



9456 Laguna Springs Drive, Elk Grove, CA, USA, Elk Grove
California 95758-7987
Mailing Address: 9456 Laguna Springs Drive, Elk Grove, CA, USA
Phone: 12312312333
Fax: +12345678901
After hour Contact: test@gmail.com

DETAIL	VALUE
Dispatcher	Test User Ultraship
Phone	+11111111111
Email	testuser@ultraship.ai
Reference ID	LD64407
Booked On	28-Jul-2026
Rate Con Date	28-Jul-2026
Pickup Date	31-Jul-2026

Carrier Details

CARRIER	MC #	PHONE	EQUIPMENT	AGREED AMOUNT (USD)	SIZE (IN FEET)
10314803 CANADA INC	MC1146347	(905) 670-6882	Flatbed	\$50.00 USD	10

Stops

#	Stops	Date/Time	Commodity			
			S. No	Commodity	Weight	Quantity
1	Pickup test automation pickup Illinois State Police, 100 W Randolph St, Chicago, IL 60601, USA PO/Container No 363028 Load Type FTL	Appt Type appointment Shipping Date & Time 07/31/2026 Appointment Time -				
			1	Ceramics	422	10
			Description Automation pickup description			
2	Drop test automation drop New York University Jeffrey S. Gould Welcome Center, 50 W 4th Street, New York, NY 10012, USA PO/Container No 816843 Load Type FTL	Appt Type fcfs Delivery Date & Time 08/02/2026 Appointment Time -				
			1	Ceramics	422	10
			Description This is automation drop location			

Rate Breakdown

Base Carrier Rate	50.00 USD
Total	50.00 USD

Shipper Instructions (Sample)

Please check in with the shipping office upon arrival. Freight must be loaded according to the scheduled appointment time. Contact the shipper immediately if any delays are expected.

Carrier Instructions (Sample)

Driver must call dispatch after pickup and provide status updates every morning. Submit signed POD within 24 hours of delivery to avoid payment delays

Driver Details

DRIVER NAME	CELL NO	TRUCK NO	TRAILER NO
Playwright Driver	(415) 555-5072	68972	24912

Accepted by	Date	Signature
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